Qualitative and quantitative situational characteristics of muscle strains in sports: a systematic review and meta-analysis

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ABSTRACT

Objectives To explore the injury-inciting situational characteristics of indirect and non-contact muscle injuries in sports, commonly referred to as muscle strains; and to investigate differences and similarities of injury patterns across muscle groups.

Data sources PubMed, Web of Science, SPORTDiscus and Google Scholar.

Eligibility criteria for selecting studies Studies investigating injury-inciting situational characteristics (among others, injury contact mechanisms, joint positions, movement directions) of indirect and non-contact muscle injuries in sports through video recordings.

Results Twenty-one studies reporting the injury-inciting situational characteristics of 728 indirect and non-contact muscle injuries were included. Non-contact mechanisms were more common (74%) than indirect contact mechanisms (26%). Most injuries were either running-related or occurred during sport-specific manoeuvres involving muscle-tendon unit length changes under active muscle contraction. For hamstring injuries, the most frequently reported injury kinematics comprised a knee joint position close to extension (underlying movement direction: flexion to extension) and a flexed hip joint position (underlying movement direction: variable). For adductor injuries, injury kinematics were characterised by rapid muscle lengthening due to hip extension, abduction and external rotation. For rectus femoris injuries, the observed injury kinematic comprised a flexing hip joint movement and extending knee joint movement. For calf injuries, the typical injury pattern comprised an ankle dorsiflexion movement with the knee being close to extension and the ankle in >10° dorsiflexion at the assumed injury time.

Conclusion This systematic review found distinct, identifiable injury patterns for indirect and non-contact muscle injuries. While similar situational characteristics were identified across injuries, some patterns were more specific to particular injury locations and sports.

PROSPERO registration number PROSPERO registration number CRD42023472252.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Indirect and non-contact muscle injuries, often referred to as muscle strains, occur in specific sporting scenarios (injury situations).
⇒ Multiple dimensions of situational characteristics, including contact mechanisms, joint positions or neuromuscular force production, have been investigated to understand causation of muscle injuries.
⇒ Despite the implementation of preventive measures, clinicians continue to face challenges in reducing the incidence of muscle injuries in sports.

WHAT THIS STUDY ADDS

⇒ Muscle injury patterns were characterised from 728 video-detected muscle injuries in sports.
⇒ Qualitative and quantitative situational characteristics (eg, injury contact mechanism, joint positions at injury time) of hamstring, adductor, quadriceps and calf muscle injuries are described in detail.
⇒ While general principles of muscle injury causation are applicable (eg, high muscle activation while being at length during the assumed time of injury), certain injury patterns are more specific to particular injury locations and sports.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Knowledge of the situational characteristics underlying muscle strains assists clinicians in accurate diagnosis and treatment decision-making.
⇒ Future preventive approaches for muscle strains should be guided by a distinct understanding of injury causation.

INTRODUCTION

Muscle injuries are common in recreational, trained and elite athletes,[1] and may represent the most common time-loss injury in sports.[2] Awareness of their high incidence has raised researchers' and clinicians' interest in developing and evaluating preventive strategies.[3] However, despite these efforts, the number of muscle injuries has not decreased in recent years.[3–5] Among various explanations, one is the difficulty of capturing the complex, multifactorial underlying contributing factors of muscle injuries.[1] To develop more efficient injury prevention programmes, a research strategy that returns to the roots and gains a deeper understanding of specific injury situations could

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prove useful.[6] Understanding the situational characteristics with the sequence of events and activities leading to an injury (eg, injury contact mechanisms, athlete's movements, joint kinematics) can guide the development of more effective prevention strategies.[6]

In order to determine injury patterns, a detailed analysis of qualitative situational characteristics (eg, injury contact mechanisms) and quantitative situational characteristics (eg, joint angles at injury occurrence) is necessary.[6] The use of systematic video analyses has been on the rise, and its potential for analysing situational characteristics and developing preventive measures has been outlined.[7–9] With the use of video analysis, it is possible to examine complex sport-specific situations leading to injury, including interactions with other players, movement patterns and body segment positions, among others. Previous video analysis studies have primarily focused on team sports such as professional football or rugby, concentrating on individual muscle groups like hamstring or adductor muscles.[10–16] With an increasing number of studies investigating different injury locations and types of sports, a systematic review and meta-analysis on the situational characteristics and patterns of muscle injuries across various sports and injury locations is warranted.

Therefore, the aim of this systematic review with meta-analysis was to outline the injury patterns of indirect and non-contact muscle injuries with underlying qualitative situational characteristics (eg, indirect contact vs non-contact, joint movement direction) and quantitative situational characteristics (eg, joint angles at assumed injury frame), and to investigate their differences and similarities across muscle groups and sports. It was hypothesised that distinct injury patterns exist regardless of the affected muscle group. Identifying these typical patterns could enhance our understanding of muscle injuries and provide a basis for developing tailored and holistic preventive measures.

METHODS

This systematic review with meta-analysis follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)[17] and Quality Appraisal for Sports Injury Video Analysis Studies (QA-SIVAS)[8] guidelines. It was prospectively registered with PROSPERO (CRD42023472252).

Search strategy and study selection

The initial systematic search was performed using PubMed, Web of Science, SPORTDiscus and Google Scholar databases in October 2023 with the following search terms:

#1 video

#2 muscle OR hamstring OR gastrocnemius OR soleus OR biceps OR triceps OR quadriceps OR rectus OR flexor OR extensor OR adductor OR abductor OR pectoralis

#3 injury

#1 AND #2 AND #3

PubMed, Web of Science and SPORTDiscus searches were performed without third-party software. The Google Scholar search was performed by using scholarly research software (Harzing's, V.8.9.4538.8589, Australia). The forward search used the aforementioned scholarly research software (Harzing's), and the backward search used Web of Science (Clarivate Analytics, Philadelphia, USA). All search results were exported to Endnote (V.21.2, Clarivate Analytics). After the removal of duplicates, the remaining records were exported to rayyan.ai.[18] Two independent reviewers (TH and LSF) screened the title, abstract and, if necessary, the full text using predefined inclusion and exclusion criteria. Inclusion criteria were: (1) situational analysis of muscle injury, (2) imaging confirmation of injury, (3) video recording of injury, (4) sport-related activities, (5) English language and (6) peer-reviewed article. Exclusion criteria were: (1) non-human participants, (2) in vitro or cadaver research, (3) injuries that occurred outside of sports-related activities, (4) other methods than video recording and (5) opinion pieces, systematic reviews and meta-analyses, editorials, commentaries, letters to the editor, conference proceedings/abstracts, book chapters and non-peer-reviewed articles. In the case of remaining disagreement, a third reviewer rated according to the implemented criteria (TG). A forward 'cited by' and backward citation search was performed for all included studies. To update the review with the most recent data, a post hoc literature search update was performed in August 2025 using the initial databases.

Quality assessment

Two independent investigators (SW and LSF) performed the quality assessment using the QA-SIVAS scale.[8] QA-SIVAS is a checklist of 18 items assessing the studies' data source, study protocol, results and discussion, among others. Each item can score 1 point, resulting in a final QA-SIVAS score of up to 18. The reliability and validity of QA-SIVAS for evaluating video analysis in sports injury research has been previously demonstrated.[8] Studies with a score <60% are considered low quality, 60%–70% moderate quality, 71%–80% good quality and >80% high quality.[8] Disagreements between the two reviewers were resolved through discussion or by a third reviewer (TG) when necessary.

Data extraction

The following data were extracted from each study by two independent investigators (TG and LSF): injury location (muscle or muscle group), video data source, number and qualification of raters, duration of data collection, country, population characteristics (type of sport, country, gender), injury rates, injury contact mechanisms, injury-inciting activities, injury kinematics (joint movement/angle at assumed injury frame) and match time of injury occurrence.

Due to missing international consensus on terminology of injury-inciting activities and movement activities, the original definitions from the respective studies were used (eg, *injury mechanisms, situational patterns, injury-inciting events, injury kinematics*) and unified as outlined in box 1. Joint kinematics were analysed according to the neutral-zero method.

Similarly, for terminology of injury location, definitions were used as reported in the studies (eg, *hamstring injuries* were reported separately from *thigh injuries*). Data on injury contact mechanisms (direct, indirect, non-contact) were extracted by using the reported numbers from each study. In line with previous research,[11 12 16 19–21] all direct contact muscle injuries were excluded from further analysis to avoid the inclusion of contusions or lacerations. Indirect and non-contact injuries were reported together for studies including both contact mechanisms. In addition, non-contact injuries were reported separately for studies that focused exclusively on non-contact injuries or when data on non-contact injuries could be extracted. According to Klein *et al*[22] and Strengthening the Reporting of Observational Studies in Epidemiology for Sport Injury and Illness Surveillance,[23] a non-contact injury is present if the injury occurred without any direct or indirect contact from another external source. A muscle injury is considered indirect if sustained through contact with other athletes or sources. Thus, force is not directly applied to the injured area, but contributing

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Box 1 Definitions and terminology on injury causation

Injury causation: combination of all factors contributing to tissue damage caused by (sports) trauma.

Situational characteristics: a situational injury characteristic refers to an attribute at or around a particular injury time (meaning that this feature can change before or after the injury, eg, within seconds). Situational characteristics can be investigated on both qualitative and quantitative levels:

⇒ Qualitative characteristics: qualitative injury characteristics refer to descriptive data. Examples of qualitative characteristics include injury contact mechanism (eg, direct, indirect, non-contact), injury-underlying activities (eg, running, jumping) or joint movements at the assumed injury time (eg, flexing, extending).

⇒ Quantitative characteristics: quantitative injury characteristics refer to numeric data. Examples include the joint angle at the assumed injury frame or the timing of injury during match play.

Injury pattern: characteristics that are repetitively found to describe a specific injury can be termed an injury pattern. For example, a water skier with a flexed hip and straight knee experiencing a forward pull may be considered a typical injury pattern for a hamstring injury.

indirectly to injury (eg, a push from the opponent player leading to perturbation and resulting situational injury pattern). A direct contact injury is present if the contact leads to the injury in an immediate manner.

In addition, we clustered the number of injuries that have been considered as being running-related (eg, running, sprinting, acceleration). Historically, muscle injuries have been considered as either running-related or stretching-related. According to Vermeulen *et al*,[21] the primary factor of stretching-related hamstring injuries is excessive stretching, involving significant lengthening, either slow (controlled movements such as splits) or fast (such as kicking, reaching) actions. Also, the kinetic chain (open vs closed) at the assumed injury frame was reviewed as suggested by Gronwald *et al*,[12] as well as the injury-inciting activities reported by the included studies. Data extraction of injury kinematics included both joint movement and joint angles at the assumed injury frame. If applicable, the time of injury occurrence during match play was assessed for team sports.

Equity, diversity and inclusion

Equity, diversity and inclusion were strategic objectives of this systematic review. The research team comprised male and female researchers of different origins, career levels and professions. Concerning data inclusion, athletes were considered eligible irrespective of geographic location, gender and sports.

Data synthesis

Meta-analytical data pooling was employed to report the proportions of injury contact mechanisms, injury-inciting activities and timing of injury. More specifically, logit transformation was applied to calculate weighted summary proportions with 95% CIs using a DerSimonian and Laird random-effects model. Analyses were conducted using RStudio (V.2024.09.1+394, Boston, Massachusetts, USA), and the meta and metafor packages as described by Wang.[24] A sensitivity analysis was undertaken for studies with good and high quality according to QA-SIVAS. Direct contact injuries were excluded from data synthesis. Case reports were not included in the meta-analysis. Studies with significant overlap in their study population and evaluated time frame were counted as one study for the purpose of data synthesis.

RESULTS

Literature search

The initial systematic searches identified 3314 records (PubMed: 1146, Web of Science: 903, SPORTDiscus: 266, Google Scholar: 999). After removal of 632 duplicates, 2682 records were screened for titles and abstracts, of which 2627 were excluded. All disagreements were resolved through discussion without the need for a third reviewer. The remaining 55 records were screened for full text, of which 13 articles met the predefined criteria.[10 12–14 19 22 25–31] Additionally, 994 articles were identified through backward (399 results) and forward (595 results) search. After removing 263 duplicates, 731 records were screened for titles and abstracts, and the full text of 36 records was assessed. Five additional studies were then included.[11 15 16 21 32] The post hoc search revealed three additional studies to be included.[20 33 34] This process resulted in a total inclusion of 21 studies. The results of the literature search are presented as a PRISMA flow chart in figure 1.

Study characteristics

Of the 21 studies reporting on 728 muscle injuries, 17 studies[10–14 16 19–22 26 28–30 32–34] investigated muscle injuries in team sports, while three studies reported on individual sports.[15 25 27] One study reported on multiple team and individual sports.[31] The majority of studies (n=13) reported on football.[10–12 14 16 19–22 26 30 33 34] The other sports investigated were basketball,[32] cricket,[29] handball,[28] rugby,[13] powerlifting (deadlifts),[15] running[27] and treadmill running.[25] Eighteen studies reported exclusively on male athletes.[10–15 19–22 25–30 32 33] Two studies investigated muscle injuries in both male and female athletes,[16 31] and one study analysed only female athletes.[34] The duration of data collection was reported in 18 out of 21 studies.[10–14 16 19–22 26 28–34] The time of data collection ranged between 1 year and 32 years (median=4 years). Detailed information about study characteristics is given in table 1.

Most studies (n=19) reported exclusively on a single muscle group.[12–16 19–22 25–34] The remaining two studies assessed multiple lower limb muscle groups.[10 11] The studies reported on obturator (n=1),[30] thigh (n=3),[22 28 32] hamstring (n=12),[10–13 15 16 21 25 26 31 33 34] adductor (n=2),[10 11] adductor longus (n=2),[14 20] quadriceps (n=2),[10 11] rectus femoris (n=1),[19] calf (n=2)[10 11] and gastrocnemius (n=2)[27 29] muscle injuries.

Thirteen studies relied on television broadcasting (with or without additional video data sources),[10–14 16 19–22 26 28 33] two studies used recordings of lab-based motion capture systems,[25 27] three studies used online and social media platforms[15 31 34] and three studies did not specify their video data source.[29 30 32]

Quality assessment

The methodological study quality was high for two studies,[21 34] good for nine studies,[11–14 16 19 20 22 26] moderate for seven studies[10 25 27 28 31–33] and low for three studies.[15 29 30] The QA-SIVAS scores ranged from 28% to 89%, with an overall mean score of 68% (SD=14%). All disagreements between the two raters were resolved by discussion without a need for a third rater.

The majority of articles (n=20) clearly stated the aim of the study,[10–16 19–22 25–28 30–34] with 71% (n=15) providing

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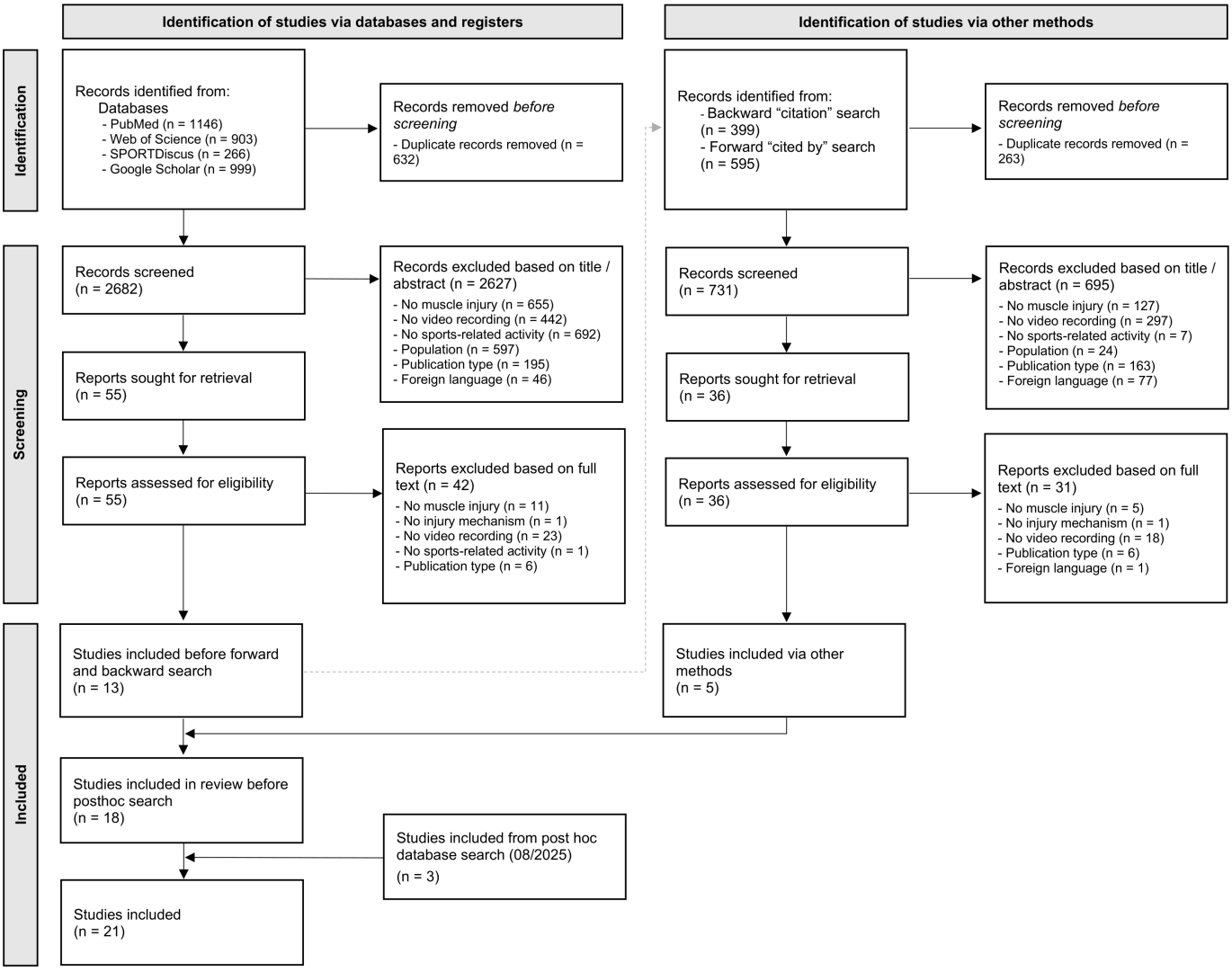

Figure 1 The Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart of the literature search.

comprehensive information about the analysed population.[10 13 14 16 19–21 25–27 29–31 33 34] In 52% of the studies (n=11),[10–13 16 21 22 28 32–34] the population was categorised as representing a homogenous group. Four studies (19%) provided detailed information about the video source and quality.[14 25 27 34] The study protocol was clearly outlined in 90% (n=19) of the studies.[10–16 19–22 25–28 31–34] A systematic process of video analysis was described in 81% (n=17) of the studies.[10–14 16 19–22 25–28 32–34] Medical information was reported in 76% of the articles (n=16).[10 12–14 19–22 25–30 32 34] Nineteen studies reported the number of raters,[10–16 19–22 25 26 28 29 31–34] and in most of these studies (n=14), two or more raters assessed the videos independently.[10–14 16 19–22 26 28 31 34] The raters' occupations and experience were reported in six studies (29%).[11 16 20 21 32 34] A comparison with a control group was made in one study.[27] While most studies analysed biomechanics, only 10% (n=2) referred to validated methods,[25 27] and 67% (n=14) inserted example figures or photographs of the injury-inciting event.[11–16 19–22 26 31 33 34] Injury incidents[10–16 19–22 25–34] and context[10–14 16 19–22 25–34] were presented in 21 studies (100%) and 20 studies (95%), respectively. A structured description of the results was provided in 95% of studies (n=20).[10–16 19–22 25–28 30–34] All studies included a discussion of other relevant trials in the field (n=21).[10–16 19–22 25–34] A discussion on how to use the study findings in clinical practice was given in 81% of studies (n=17)[11–16 19–22 26 28 30–34] of all studies. Limitations were reported in 90% the studies (n=19).[10–16 19–22 25 26 28 30–34] The quality assessment using the QA-SIVAS scale is presented in online supplemental file 1.

Situational injury characteristics: contact mechanisms

The 21 studies reported on 787 muscle injuries (non-contact: 583, indirect contact: 145, direct contact: 29, not specified: 30). One study did not specify the contact mechanism and was excluded from further analysis.[30] In addition, one study could not specify the contact mechanism for one injury out of 63 injuries; the unspecified injury was excluded in our analysis apart from the analysis of the timing of injury.[21] Most studies (n=14)[11–16 19–22 26 31 32 34] reported on both indirect and non-contact muscle injuries, while six studies focused exclusively on non-contact injuries.[10 25 27–29 33] The meta-analytical pooling revealed that non-contact muscle injuries occurred in 74% of the cases (95% CI 67% to 79%, I^2=49%) when considering studies that included both indirect and non-contact injuries.

The subgroup data pooling for specific muscle groups is presented in figure 2. For thigh injuries (two studies), non-contact injuries accounted for 64% (95% CI 53% to 73%). For hamstring injuries (nine studies),[11–13 15 16 21 26 31 34] non-contact

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Table 1 Characteristics of included studies

Study	Injury location	Sport	Country	Cohort	N (f)	Years included
Achenbach *et al*[32]	Thigh	Basketball	Germany	Professional basketball players	19 (0)	2014–2019, excluding season 2017/2018
Aiello *et al*[10]	Hamstring, adductor, calf, quadriceps	Football	Italy	Professional football players (Serie A)	29 (0)	2019–2022
Dao *et al*[15]	Hamstring	Power lifting (deadlifting)	n.s.	n.s.	16 (0)	n.s.
Della Villa *et al*[11]	Hamstring, adductor, calf, quadriceps	Football	Italy	Professional football players (Serie A)	103 (0)	2018–2021
Della Villa *et al*[16]	Hamstring	Football	Italy	Professional football players (Serie A)	129 (64)	2018–2021, 2017–2023
Gandarias-Madariaga *et al*[33]	Hamstring	Football	Spain	Professional football players (LaLiga)	78 (0)	2017–2018 and 2018–2019
Gronwald *et al*[12]	Hamstring	Football	Germany	Professional football players (Bundesliga 1 and 2)	52 (0)	2014–2019, excluding season 2017–2018
Heiderscheidt *et al*[25]	Hamstring	Treadmill running	n.s.	Professional skier	1 (0)	n.s.
Jokela *et al*[19]	Rectus femoris	Football	Finland and Spain	Professional football players	20 (0)	2014–2019, excluding season 2017–2018
Jokela *et al*[26]	Hamstring	Football	Finland and Spain	Professional football players	14 (0)	2017–2022
Jokela *et al*[20]	Adductor longus	Football	n.s.	Professional football players	20 (0)	2017–2023
Kerin *et al*[13]	Hamstring	Rugby	Ireland	Professional rugby players	17 (0)	2015–2018
Klein *et al*[22]	Thigh	Football	Germany	Professional football players (Bundesliga 1 and 2)	81 (0)	2014–2017
Kong[27]	Gastrocnemius	Running	USA	Kinesiology student	1 (0)	n.s.
Luig *et al*[28]	Thigh	Handball	Germany	Professional handball players (German first or second league)	31 (0)	2010–2017, excluding season 2013–2014
Orchard *et al*[29]	Gastrocnemius	Cricket	UK	Professional cricket player	1 (0)	2001
Pellegrini *et al*[34]	Hamstring	Football	Various (USA, Italy, Spain, England, France, Germany)	Professional football players	57 (57)	2017–2024
Serner *et al*[14]	Adductor longus	Football	Qatar	Professional football players	17 (0)	2013–2015
Vermeulen *et al*[21]	Hamstring	Football	Qatar	Professional football players	63 (0)	2013–2020
Wong-On *et al*[30]	Obturator externus and internus	Football	Spain	Professional football players	16 (0)	Four seasons
Yüce *et al*[31]	Hamstring	Multiple	n.s.	Elite athletes	61 (3)	1989–2021

f, female ; N, number of participants; n.s., not specified.

injuries were reported for 75% (95% CI 65% to 82%). Here, the sensitivity analysis revealed similar CI ranges. The heterogeneity was reduced from 49% to 35% and (in the subgroup of hamstring injuries) from 63% to 44%.

Situational injury characteristics: running-related injuries

For injuries characterised as running-related injuries, further qualitative and quantitative analysis was performed. Nine studies with combined reporting of indirect and non-contact injuries were included in the clustering analysis of running-related injuries (figure 3A).[11–14 19–21 26 34] The meta-analytical pooling revealed that running-related injuries accounted for 36% of the cases (95% CI 23% to 51%) with considerable heterogeneity (I^2=80%).

For thigh injuries, 45 running-related injuries were reported (114 injuries in total). Here, Klein *et al*[22] further differentiated between sprinting injuries and running injuries (32 sprinting vs 11 running). For hamstring injuries, running-related injuries represented 52% of all cases (95% CI 37% to 66%). Here, Gronwald *et al*[12] further differentiated acceleration injuries (n=14, 58%) vs high-speed running injuries (n=10, 42%). For adductor longus injuries, running-related injuries accounted for 11% of cases (95% CI 3% to 32%).

When exclusively considering non-contact injuries (figure 3B),[10–15 19 26 33 34] running-related injuries accounted for 46% (95% CI 31% to 62%), with substantial heterogeneity reported across studies (I^2=74%). In the subgroup of hamstring injuries, running-related injuries represented 61% of cases (95% CI 44% to 76%). Here, Aiello *et al*[10] differentiated between linear running (n=15, 94%) and curved running (n=1, 6%). In addition, Gronwald *et al*[12] reported acceleration injuries (n=12, 57%) vs high-speed running injuries (n=9, 43%). Running-related injuries were present in 36% of the included cases (95% CI 1% to 98%) for adductor injuries, 17% (95% CI 3% to 54%) for quadriceps injuries and 52% (95% CI 17% to 85%) for calf injuries. In the subgroup of hamstring injuries, the sensitivity analysis reduced heterogeneity by 60% (reduction of the CI by 10%). For the other muscle groups, only one study would have met the criteria for sensitivity analysis; no further meta-analytical approach was possible.

Situational injury characteristics: kinetic chain

Regarding the kinetic chain, data were extracted from nine studies on indirect and non-contact injuries.[10 11 14 15 20 27 29 31 34] Of these, 58 injuries (43%) were classified as open kinetic chain and 76 injuries (57%) were categorised as closed kinetic chain. Sixteen hamstring injuries while deadlifting[15] and 33 other hamstring injuries were classified as closed kinetic chain injuries (52%). The remaining 46 hamstring injuries were reported as open kinetic chain injuries during the late swing phase of the

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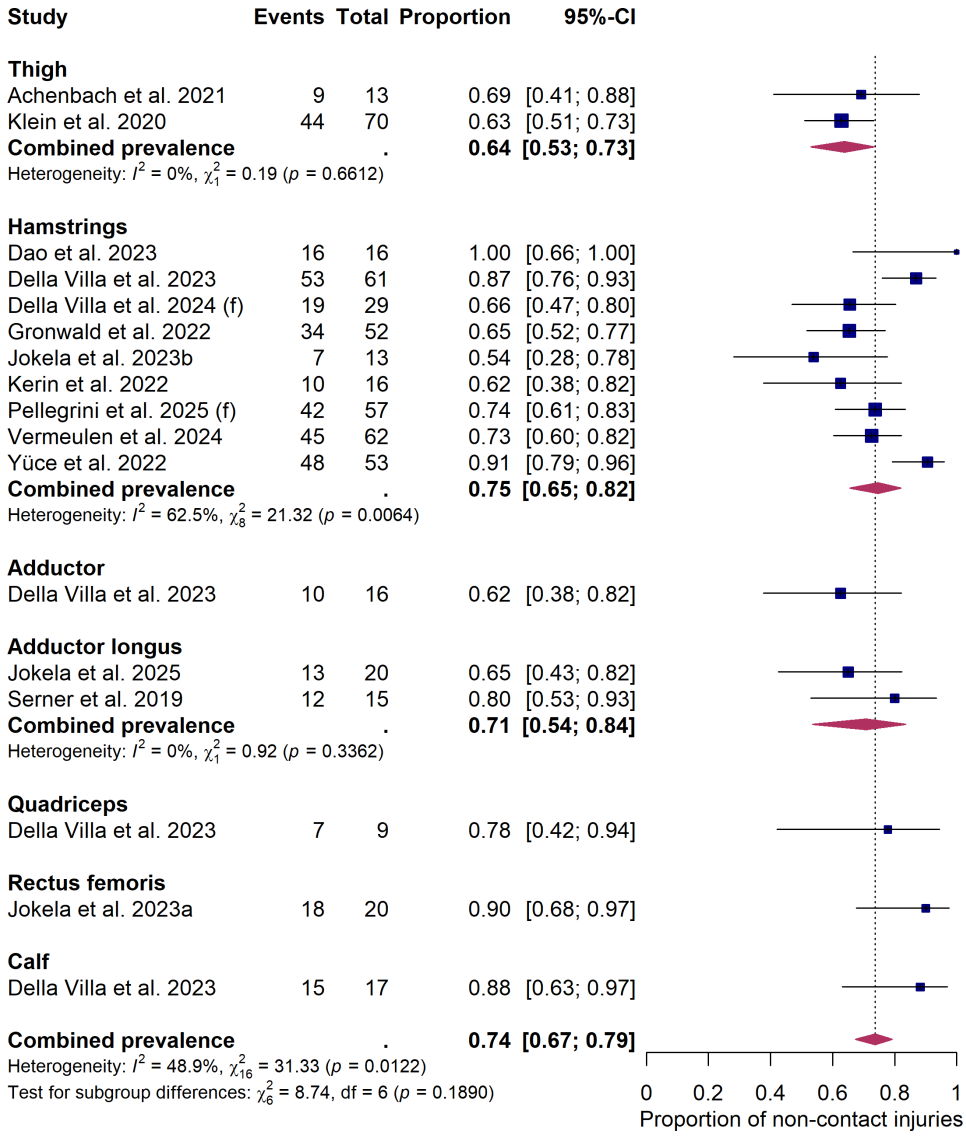

Figure 2 Proportions of non-contact muscle injuries versus indirect contact muscle injuries across muscle groups; (f) indicates female participants. The male population of Della Villa *et al* (2024) was excluded due to an overlap in study population with Della Villa *et al* (2023).

running gait cycle (48%).[11 31 34] Of these, 29 injuries belonged to a study that exclusively analysed sprinting-related injuries (n=32).[31] In addition, 10 adductor longus (32%) and two quadriceps injuries (50%) were reported as open kinetic chain injuries.[10 14 20] In contrast, 21 adductor longus (68%), two quadriceps (50%), two calf (100%) and two gastrocnemius injuries (100%) were reported as closed kinetic chain injuries.[11 14 20 27 29]

Situational injury characteristics: injury-inciting activities

With regard to injury-inciting activities, among non-contact muscle injuries, 47% were associated with running (n=144) and 21% happened while kicking (n=64). In the subgroup of football injuries, 49% were associated with running (n=134) and 23% happened while kicking (n=63). For indirect contact muscle injuries, 24% were associated with lunging (n=21) and 22% were related to running (n=20). In the subgroup of football injuries, 27% were associated with lunging (n=21) and 24% happened during running activities (n=19). By injury locations, 68% of the non-contact thigh injuries (n=36) and 30% of the indirect contact thigh injuries (n=9) happened during running activities. Non-contact hamstring injuries were most commonly associated with running, accounting for 53% of cases (n=94).[10–13 15 26] For indirect contact hamstring injuries, lunging injuries were commonly reported (n=13, 30%).[11–13 26 34] Adductor injuries frequently appeared while kicking (n=7, 32%) and reaching (n=4, 18%).[10 11] For the adductor longus, kicking (n=5, 33%), change of direction (n=5, 33%) and reaching (n=3, 20%) were the primary injury-inciting activities.[14] Studies that assessed quadriceps injuries or rectus femoris injuries reported kicking as the most common injury pattern, accounting for 64% ($n_{Quadriceps}$=7) and 80% ($n_{Rectus\ femoris}$=16) of all cases, respectively.[10 11 19] Calf injuries occurred mostly while running (n=11, 52%) or overstretching (n=7, 33%).[10 11] One gastrocnemius injury occurred while running and the other injury occurred while setting off to take a run.[27 29] A summary of common injury-inciting activities is summarised in table 2.

Situational injury characteristics: kinematics

Injury kinematics were analysed in hamstring injuries (seven studies),[12 13 15 25 26 31 34] adductor longus injuries (two studies),[14 20] rectus femoris injuries (one study)[19] and gastrocnemius injuries (two studies).[27 29] Nine studies described the joint

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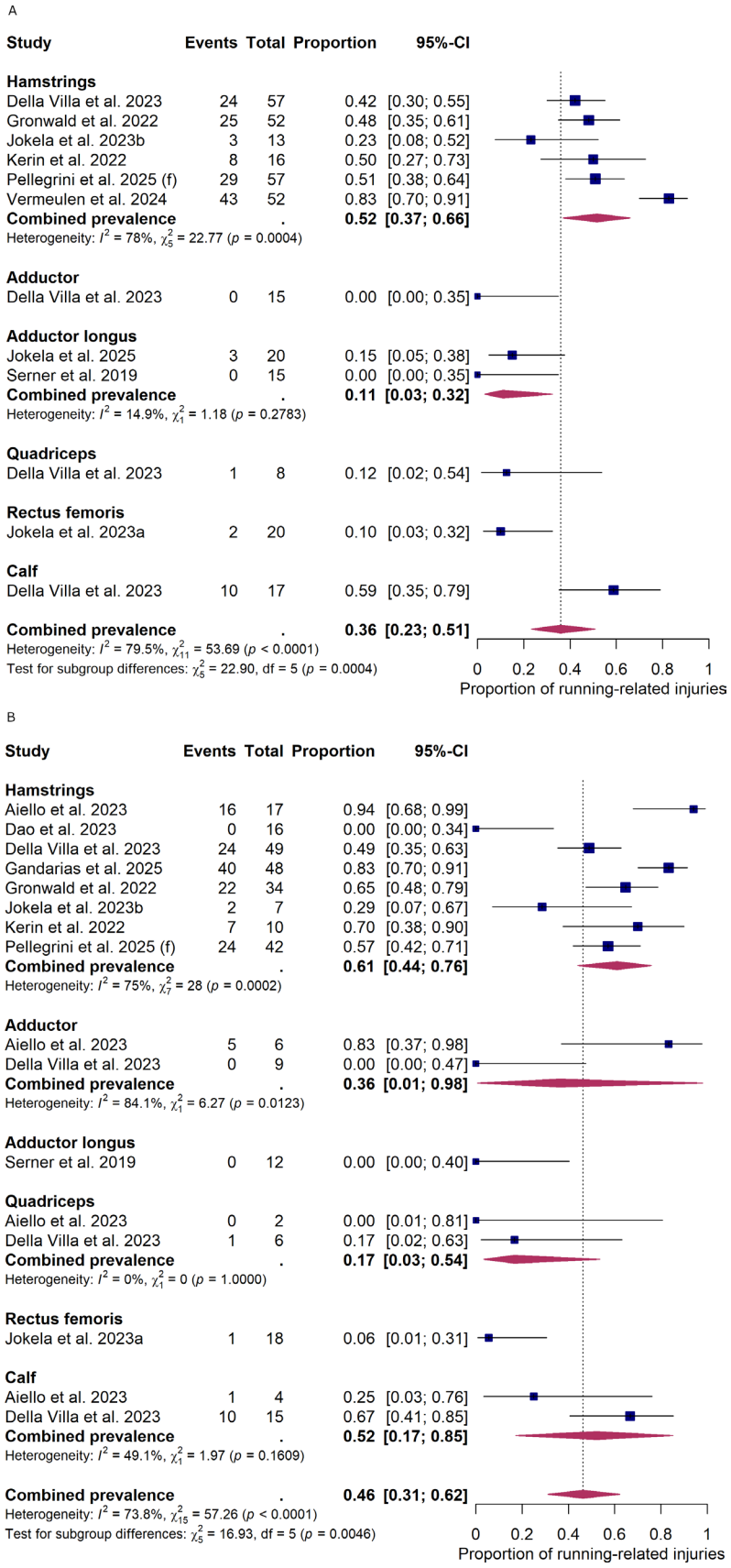

Figure 3 Proportions of running-related injuries. The first forest plot (A) refers to studies with combined reporting of indirect and non-contact injuries. The second forest plot (B) refers to non-contact injuries only; (f) indicates female participants.

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Table 2 Most common injury-inciting activities based on injury location

Injury location	Study	N	Top injury-inciting activities Non-contact injuries	Top injury-inciting activities Indirect contact injuries
Thigh	Achenbach *et al*[32]	13	Other (n=5), sprinting (n=2), stopping (n=2)	Not specified (n=4)
	Klein *et al*[22]	70	Sprinting (n=23), running (n=11), not specified (n=10)	Sprinting (n=9), lunging (n=8), not specified (n=9)
	Luig *et al*[28]	31	Sprinting (N/A), running (N/A), change of direction (N/A)	
Hamstring	Aiello *et al**[10]	17	Linear running (n=15), curved running (n=1), jumping (n=1)	
	Dao *et al*[15]	16	Deadlifting (n=16)	
	Della Villa *et al**[11]	61	Running (n=24), open kinetic chain stretching (n=13), closed kinetic chain stretching (n=6), kicking (n=6), other (n=4)	Closed kinetic chain stretching (n=6), open kinetic chain stretching (n=2)
	Gronwald *et al*[12]	52	Acceleration (n=12), high-speed running (n=9), kicking (n=7), braking or stopping: lunging (n=3), braking or stopping: landing (n=1), other (n=1), unclear (n=1)	Braking or stopping: lunging (n=13), acceleration (n=2), high-speed running (n=1), braking or stopping: landing (n=1), kicking (n=1)
	Jokela *et al*[26]	13	Running (n=2), change of direction (n=2), kicking (n=2), approach to kick (n=1)	Change of direction (n=2), running (n=1), kicking (n=1), jumping (n=1), reaching for ball (n=1)
	Kerin *et al*[13]	16	Running (n=7), kicking (n=1), stopping (n=1), turning (n=1)	Contesting ruck (n=3), running (n=1), deceleration (n=1), tackle (n=1)
	Pellegrini *et al*[34]	57	Run-type (n=24), kick-type (n=13), duel-type (n=5)	Run-type (n=5), landing from a wall jump (n=2), kick-type (n=1), duel-type (n=7)
Adductor	Aiello *et al**[10]	6	Receiving (n=2), kicking (n=2), dribbling (n=1), ball contended (n=1)	
	Della Villa *et al**[11]	16	Kicking (n=4), open kinetic chain stretching (n=3), reaching (n=1), jumping (n=1), other (n=1)	Reaching (n=3), open kinetic chain stretching (n=1), kicking (n=1), jumping (n=1)
Adductor longus	Serner *et al*[14]	15	Kicking (n=5), change of direction (n=4), jumping (n=2), reaching (n=1)	Reaching (n=2), change of direction (n=1)
Quadriceps	Aiello *et al**[10]	2	Kicking (n=2)	
	Della Villa *et al**[11]	9	Kicking (n=5), running (n=1), other (n=1)	Closed kinetic chain stretching (n=2)
Rectus femoris	Jokela *et al*[19]	20	Kicking (n=16), running (n=1), change of direction (n=1)	Running (n=1), change of direction (n=1)
Calf	Aiello *et al**[10]	4	Running (n=1), jumping (n=1), ball receiving (n=1), kicking (n=1)	
	Della Villa *et al**[11]	17	Running (n=10), closed kinetic chain stretching (n=5)	Closed kinetic chain stretching (n=2)
Gastrocnemius	Kong[27]	1	Running (n=1)	
	Orchard *et al*[29]	1	Setting off to take a run (n=1)	

*Studies with overlap in study population.
N/A, not applicable.

movement directions at the assumed injury frame (hamstring: five studies,[12 13 15 25 34] adductor longus: two studies,[14 20] rectus femoris: one study,[19] gastrocnemius: one study).[29] Additionally, all seven studies on hamstring injuries, both studies on the adductor longus injuries and both studies on gastrocnemius injuries described joint positions at the assumed injury frame.[12–15 20 25–27 29 31 34]

Regarding all injuries, the movement directions of the hip were evenly distributed, with 37% extending (n=34), 55% flexing (n=51) and 8% static (n=7). The knee was primarily extending with 79% (n=60), 5% flexing (n=4) and 3% static (n=2).

For hamstring injuries, the sagittal hip movement was extending (n=29, 30%), flexing (n=37, 38%), remaining static (n=7, 7%) or was not specified (n=24, 25%). The knee was predominantly extending (n=79, 81%), remained in a static position (n=2, 2%) or was not specified (n=16, 16%). Trunk orientation was reported in one study, showing no trend towards a specific position (flexion/neutral/extension).[12] In one study focusing exclusively on indirect contact injuries, the knee was extending (n=6), while the hip was extending in two cases and flexing in four cases.[13]

For adductor injuries, the hip movement was flexing (n=22, 76%) or extending (n=7, 24%), adducting (n=18, 69%) or abducting (n=8, 31%) and primarily externally rotating (n=7, 88%). The trunk orientation was neutral (n=7, 50%), flexed (n=3, 21%) or extended (n=4, 29%). The hip was extended (n=21, 68%), flexed (n=9, 29%) or neutral (n=1, 3%), abducted (n=26, 87%) or neutral (n=4, 13%), externally rotated (n=22, 76%), internally rotated (n=1, 3%) or in a neutral position (n=6, 21%). The knee flexion was lower than 45° in 81% of the injuries (n=26).[14 20] For rectus femoris injuries, the hip was primarily flexing (n=14, 78%) and occasionally extending (n=3, 17%). The knee was moving from flexion to extension in 14 cases (78%) and from extension to flexion in three cases (17%). The joint positions at assumed injury frame were not described in this study.[19]

One case study of a non-contact gastrocnemius injury described the movement directions at the assumed injury frame. Here, the knee was reported to be flexing and the ankle to be dorsiflexing.[29] Both case reports on non-contact gastrocnemius injuries analysed joint positions in the sagittal plane.[27 29] In both cases, the knee joint was positioned between 0° and 6°, and the ankle ranged between 10° and 20° dorsiflexion.

A detailed description of the joint configurations at the assumed injury frame is provided in table 3.

Time of injury occurrence

Injury timing during match play was analysed in five studies for indirect and non-contact injuries.[12 13 16 21 34] One study analysed injuries in male and female soccer players[16] and one study analysed only female soccer players,[34] while the remaining studies reported exclusively on male athletes. All studies reported on

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Table 3 Three-dimensional plane movement directions and joint positions at the injury frame

Injury location	Study	N	Three-dimensional plane kinematics at injury frame *Movement directions*	Three-dimensional plane kinematics at injury frame *Joint position*
Hamstring	Dao *et al*[15]	16	Non-contact injuries: Hip (sagittal) Static (7/16; 44%), flexion to extension (9/16; 56%) Knee (sagittal) Static (2/16; 13%), flexion to extension (14/16; 88%)	Non-contact injuries: Hip (sagittal) Flexed ≥90° (11/16; 69%), semi-flexed <90° (5/16; 31%), extended (0/16; 0%) Knee (sagittal) Flexed ≥90° (0/16; 0%), semi-flexed <90° (16/16; 100%), extended (0/16; 0%)
	Gronwald *et al*[12]	52	Non-contact and indirect contact injuries (stretching-related injuries, n=25): Hip (sagittal) Extension to flexion (18/25; 72%), flexion to extension (7/25; 28%) Knee (sagittal) Extension to flexion (0/25; 0%), flexion to extension (25/25; 100%)	Non-contact and indirect contact injuries (stretching-related injuries, n=25): Hip (sagittal) Flexion <45° (13/25; 52%), 45°–90° (10/25; 40%), >90° (2/25; 8%) Knee (sagittal) Flexion <45° (25/25; 100%), 45°–90° (0/25; 0%), >90° (0/25; 0%) Trunk (sagittal) Extended (8/25; 32%), flexed (10/25; 40%), neutral (7/25; 28%)
	Heiderscheidt *et al*[25]	1	Non-contact injury: Hip (sagittal) Flexion to extension (1/1; 100%) Knee (sagittal) Flexion to extension (1/1; 100%)	Non-contact injury: Hip (sagittal) Flexion 69° Knee (sagittal) Flexion 58°
	Jokela *et al*[26]	13		Non-contact injury: Hip (sagittal) Flexion <45° (4/7; 57%), 45°–90° (3/7; 43%), >90° (0/7; 0%) Knee (sagittal) Flexion <45° (7/7; 100%), 45°–90° (0/7; 0%), >90° (0/7; 0%) Trunk (sagittal) Flexion <45° (3/7; 43%), 45°–90° (0/7; 0%), >90° (1/7; 14%), neutral (3/7; 43%) Indirect contact injury: Hip (sagittal) Flexion <45° (0/6; 0%), 45°–90° (4/6; 67%), >90° (2/6; 33%) Knee (sagittal) Flexion <45° (5/6; 83%), 45°–90° (1/6; 17%), >90° (0/6; 0%) Trunk (sagittal) Flexion <45° (1/6; 17%), 45°–90° (2/6; 33%), >90° (0/6; 0%), neutral (3/6; 50%)
	Kerin *et al*[13]	16	Non-contact injuries: Hip (sagittal) Extending (10/10; 100%), flexing (0/10; 0%) Knee (sagittal) Extending (10/10; 100%), flexing (0/10; 0%) Indirect contact injuries: Hip (sagittal) Extending (2/6; 33%), flexing (4/6; 67%) Knee (sagittal) Extending (6/6; 100%), flexing (0/6; 0%)	Non-contact injuries: Hip (sagittal) Flexion (9/10; 90%), extension (1/10; 10%), neutral (0/10; 0%) Hip (coronal) Abduction (1/10; 10%), adduction (0/10; 0%), neutral (9/10; 90%) Hip (transverse) Internal rotation (1/10; 10%), external rotation (4/10; 40%), neutral (5/10; 50%) Knee (sagittal) Flexion (2/10; 20%), extension (8/10; 80%) Trunk (sagittal) Flexion (10/10; 100%), extension (0/10; 0%), neutral (0/10; 0%) Trunk (coronal) Ipsilateral (1/10; 10%), contralateral (2/10; 20%), neutral (7/10; 70%) Trunk (transverse) Ipsilateral (6/10; 60%), contralateral (1/10; 10%), neutral (3/10; 30%) Indirect contact injuries: Hip (sagittal) Flexion (6/6; 100%), extension (0/6; 0%), neutral (0/6; 0%) Hip (coronal) Abduction (2/6; 33%), adduction (0/6; 0%), neutral (3/6; 50%), unclear (1/6; 17%) Hip (transverse) Internal rotation (0/6; 0%), external rotation (2/6; 33%), neutral (3/6; 50%), unclear (1/6; 17%) Knee (sagittal) Flexion (2/6; 33%), extension (4/6; 67%) Trunk (sagittal) Flexion (6/6; 100%), extension (0/6; 0%), neutral (0/6; 0%) Trunk (coronal) Ipsilateral (3/6; 50%), contralateral (0/6; 0%), neutral (3/6; 50%) Trunk (transverse) Ipsilateral (2/6; 33%), contralateral (1/6; 17%), neutral (2/6; 33%), unclear (1/6; 17%)
	Pellegrini *et al*[34]	57	Non-contact and indirect contact injuries (run-related, kick-related and duel-related injuries, n=39): Hip (sagittal) Extension to flexion (15/39; 39%), flexion to extension (not specified) Knee (sagittal) Extension to flexion (not specified), flexion to extension (23/39; 59%)	Non-contact and indirect contact injuries: Hip (sagittal) Flexion 0°–30° (11/39; 28%), 30°–60° (10/39; 26%), 60°–90° (5/39; 13%), extension 0°–30° (8/39; 21%), not specified (5/39; 13%) Knee (sagittal) Flexion 0°–30° (23/39; 59%), 30°–60° (4/39; 10%), 60°–90° (7/39; 18%), not specified (5/39; 13%) Trunk (sagittal) Extended (5/39; 13%), flexed (16/39; 41%), neutral (18/39; 46%)

Continued

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Table 3 Continued

Injury location	Study	N	Three-dimensional plane kinematics at injury frame *Movement directions*	Three-dimensional plane kinematics at injury frame *Joint position*
	Yüce *et al*[31]	53		Non-contact and indirect contact injuries (sprinting-type injuries, n= 32) Trunk (sagittal) Flexion (18/32; 56%), extension (14/32; 44%) Trunk (coronal) Neutral (22/32; 69%), ipsilateral (4/32; 13%), contralateral (6/32; 19%) Trunk (transverse) Neutral (23/32; 72%), ipsilateral (3/32; 9%), contralateral (6/32; 19%) Thigh (sagittal) Flexion (23/32; 72%), extension (9/32; 28%) Knee (sagittal) Flexion (30/32; 94%), extension (2/32; 6%) Ankle (sagittal) Neutral (14/32; 44%), flexion (16/32; 50%), extension (2/32; 6%)
Adductor longus	Jokela *et al*	20	Non-contact and indirect contact injuries: Hip (sagittal) Extension to flexion (15/17; 88%), abduction to adduction (14/17; 82%)	Non-contact and indirect contact injuries: Hip (sagittal) Flexed (5/17; 29%), extended (12/17; 71%) Hip (frontal) Abducted (15/17; 88%), neutral (2/17; 12%) Hip (transverse) External rotation (15/17; 88%), neutral (2/17; 12%) Knee (sagittal) Flexed <45° (16/17; 94%), 45°–90° (1/17; 6%) Trunk (sagittal) Flexed (9/17; 53%), extended (3/17; 18%), neutral (5/17; 29%) Trunk (frontal) Towards (8/17; 47%), away (2/17; 12%), neutral (7/17; 41%) Trunk (transverse) Towards (3/17; 18%), away (7/17; 41%), neutral (7/17; 41%)
	Serner *et al*[14]	15	Non-contact injuries: Hip (sagittal) Extension to flexion (5/6; 83%), extension (3/4; 75%) Hip (coronal) Abduction to adduction (4/4; 100%), abduction (3/3; 100%) Hip (transverse) Externally rotated (6/7; 86%) Indirect contact injuries: Hip (sagittal) Extension (1/2; 50%) Hip (coronal) Abduction (2/2; 100%) Hip (transverse) Externally rotated (1/1; 100%)	Non-contact injuries: Trunk (sagittal) Flexion (1/12; 8%), neutral (7/12; 58%), extension (4/12; 33%) Trunk (coronal) Flexion to injury side (3/11; 27%), neutral (7/11; 64%), away from injury side (1/11; 9%) Trunk (transverse) Towards injury side (1/11; 9%), neutral (3/11; 27%), away from injury side (7/11; 64%) Pelvis (transverse) Posterior rotation of injured side (10/12; 83%), neutral (2/12; 17%) Hip (sagittal) Flexion (3/11; 27%), extension (8/11; 73%) Hip (coronal) Abduction (9/11; 82%), neutral (2/11; 18%); Hip (transverse) External (6/10; 60%), neutral (3/10; 30%), internal (1/10; 10%) Knee (sagittal) Flexion <45° (8/12; 67%), 45°–90° (4/12; 33%) Indirect contact injuries: Trunk (sagittal) Flexion (2/2; 100%) Trunk (coronal) Flexion to injury side (1/2; 50%), away from injury side (1/2; 50%) Trunk (transverse) Towards injury side (2/3; 67%), away from injury side (1/3; 33%) Pelvis (transverse) Posterior rotation of injured side (2/2; 100%) Hip (sagittal) Flexion (1/3; 33%), extension (1/3; 33%), neutral (1/3; 33%) Hip (coronal) Abduction (2/2; 100%) Hip (transverse) External (1/2; 50%), neutral (1/2; 50%) Knee (sagittal) Flexion <45° (2/3; 67%), 45°–90° (1/3; 33%)
Rectus femoris	Jokela *et al*[19]	20	Non-contact and indirect contact injuries: Hip (sagittal) Extension to flexion (14/18; 78%), flexion to extension (3/18; 17%) Knee (sagittal) Extension to flexion (3/18; 17%), flexion to extension (14/18; 78%)	
Gastrocnemius	Kong[27]	1		Non-contact injury: Angles and moments at touchdown Hip sagittal angle 57° Knee sagittal angle 6° Ankle sagittal angle 20.7° Hip sagittal joint moment 28 Nm Knee sagittal joint moment −38 Nm Ankle sagittal joint moment 8 Nm
	Orchard *et al*[29]	1	Non-contact injury: Knee joint very slowly flexing or no velocity Ankle joint very slowly dorsiflexing	Non-contact injury: Ankle: 10°–15° dorsiflexion Knee: 0°–5° flexion Forefoot loading

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hamstring injuries. Overall, the meta-analytical pooling revealed that 54% (95% CI 46% to 62%) of the injuries occurred in the first half. For female athletes, the proportion was 51% (95% CI 41% to 62%). The heterogeneity of all studies was substantial (I^2=41%). Here, the sensitivity analysis revealed no changes as all included studies met the criteria to be part of the sensitivity analysis.

For non-contact injuries specifically, data were available from two studies,[10 13] one of which reported on four different muscle groups (hamstring, adductor, quadriceps, calf).[10] The proportion of injuries in the first half was 53% (95% CI 37% to 69%). Indirect contact injuries were not analysed due to limited data. The forest plot of the injury proportions of injuries in the first half is provided in figure 4. Here, the sensitivity analysis excluded one of the two studies. Therefore, meta-analysis was not possible.

DISCUSSION

The present analysis of 728 video-detected indirect and non-contact muscle injuries identified distinct injury patterns based on injury location and type of sport. Non-contact injury mechanisms were more prevalent than indirect contact mechanisms (74% vs 26%). The most common injury-inciting activity was running, while most of the remaining injuries occurred

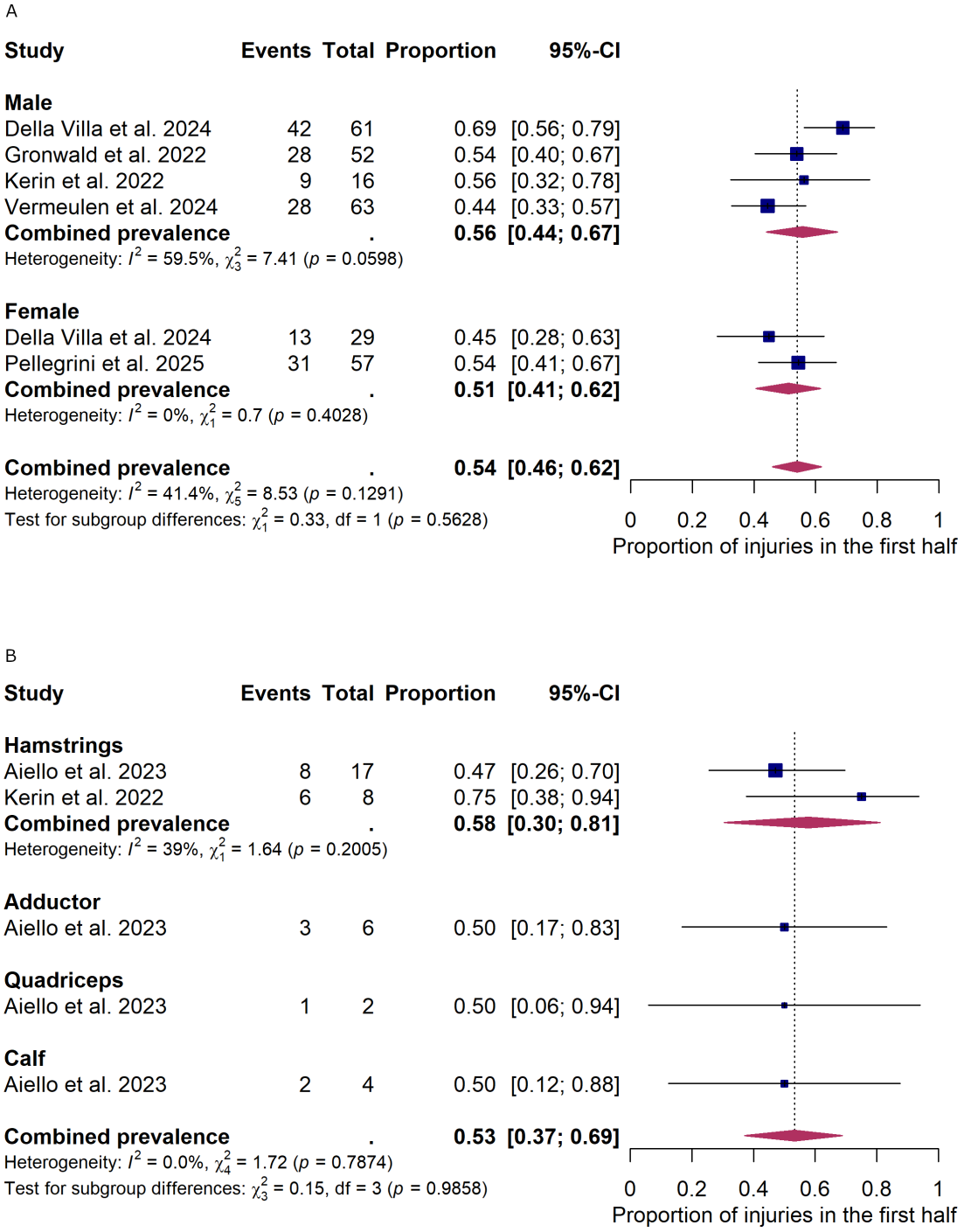

Figure 4 Proportions of injuries in the first half versus second half. The first forest plot (A) refers to studies with combined reporting of indirect and non-contact injuries. The second forest plot (B) refers to non-contact injuries.

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during sport-specific manoeuvres involving rapid changes in muscle-tendon unit length under active muscle contraction. For hamstring injuries, the most frequently observed kinematic injury pattern involved an extending knee joint movement with a joint angle of <45° at the assumed injury time while the hip joint was in a flexed position; there was no consistent position for the trunk. For the adductor muscles, injuries were characterised by rapid muscle lengthening due to hip extension, hip abduction and hip external rotation. For rectus femoris injuries, the most frequently observed kinematic injury pattern involved a flexing hip joint movement and extending knee joint movement (eg, kicking manoeuvre). For calf injuries, the most frequently observed kinematic injury pattern involved an ankle dorsiflexion movement with joint angles of <10° knee joint flexion and approximately 10°–20° ankle dorsiflexion at the assumed moment of injury. A summary of the most common injury patterns for hamstring, adductor, quadriceps and calf injuries is illustrated in figure 5.

Hamstring injuries

Three-quarters of the hamstring injuries were non-contact, while one-quarter resulted from indirect contact injuries. Consistent with this finding, previous reports have identified

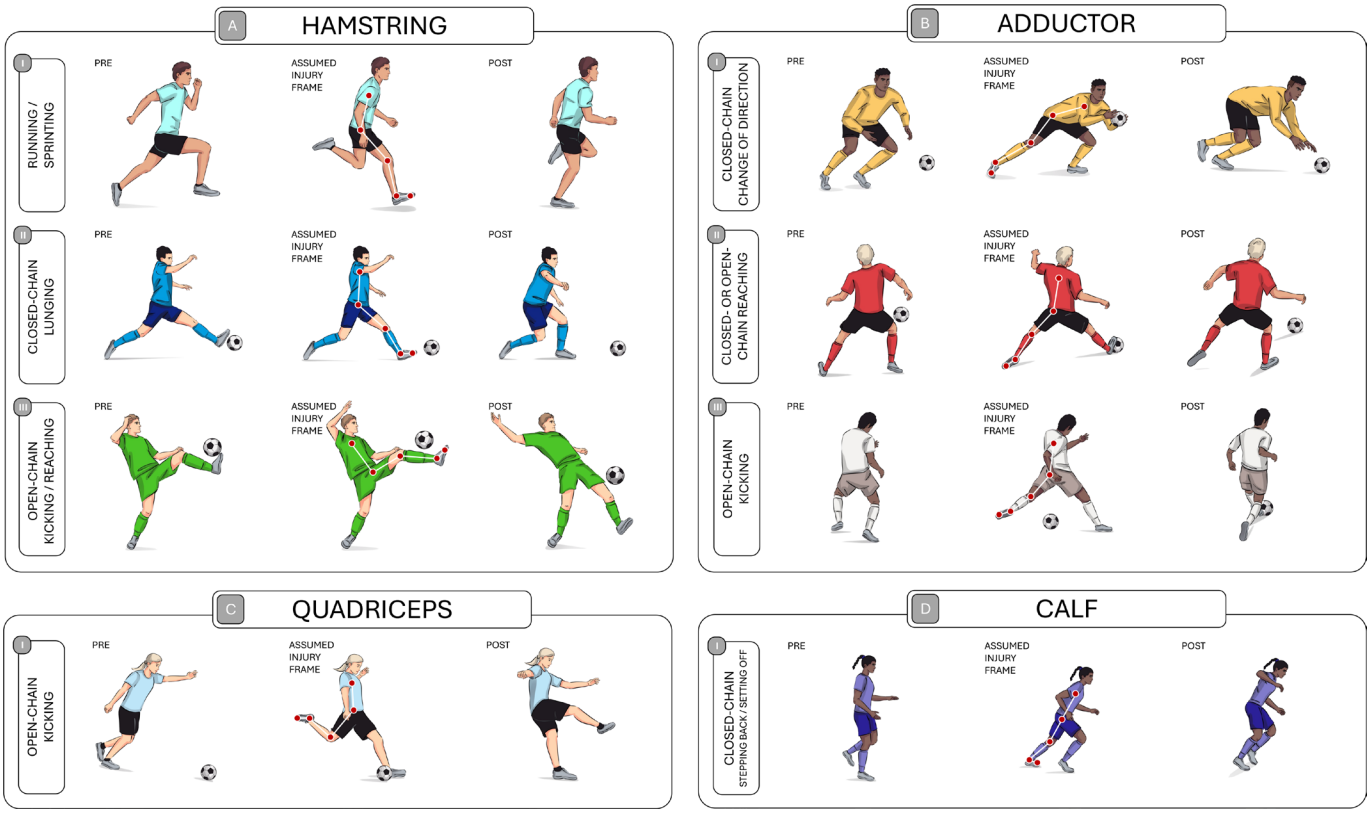

Figure 5 Illustrations of situational patterns for indirect and non-contact muscle injuries (muscle strains). Please note that only a selection of the most common injury patterns based on included studies is illustrated (acknowledging the predominance of studies investigating male football players). (A-I) Hamstring (running/sprinting): hamstring injuries are frequently seen during high-speed running or acceleration phases. Modelling studies and case reports identified the open-chain late swing phase as being most vulnerable to injury. During this phase of the gait cycle, the muscle-tendon unit of the biceps femoris lengthens. However, confirmation of this finding appraising systematic real-world video data is yet to be done, and the specific running phase in which athletes are most vulnerable to hamstring injury remains a matter of debate. (A-II) Hamstring (closed kinetic chain lunging injury pattern): the athlete performs a decelerating closed kinetic chain manoeuvre. At the assumed time of injury, the knee joint is close to full extension, the hip joint is in a flexed position (ie, lunging position). (A-III) Hamstring (open-chain kicking or reaching injury pattern): open-chain injury patterns are typically observed during kicking or reaching manoeuvres. Injury kinematics comprise a flexed hip joint combined with an extending knee joint movement. (B-I) Adductor (closed-chain change of direction injury pattern): changes of directions are common situational patterns for adductor muscle injuries. The athlete performs a change of direction to catch a ball opposite to the moving direction. At the assumed time of injury, the injured leg is abducted and externally rotated while the adductor muscles are simultaneously activated to perform the deceleration and change of direction manoeuvre. (B-II) Adductor (closed-chain or open-chain reaching injury pattern): the athlete performs a reaching manoeuvre with the non-injured leg towards the ball. At the assumed time of injury, the adductor muscle-tendon unit of the injury leg is lengthening due to hip extension, hip abduction and hip external rotation. (B-III) Adductor (open-chain kicking injury pattern): this injury pattern shows similar injury kinematics (including hip abduction and external rotation) but is an open-chain injury pattern due to the player's intention of kicking a ball with the injury-sided leg. (C–I) Quadriceps (open-chain kicking injury pattern): a commonly observed injury kinematic of quadriceps injuries comprises a flexing hip joint and extending knee joint movement (ie, kicking manoeuvre). (D-I) Calf (closed-chain stepping back injury pattern): in the illustrated example, the athlete is setting off to take a run (eg, by performing a back-step manoeuvre). These manoeuvres are seen in running or football and are common in racquet sports (leading gastrocnemius muscle injury to be named 'tennis leg') or basketball. The underlying joint movements are ankle dorsiflexion and knee extension, thereby lengthening the calf muscle tendon unit. At the assumed time of injury, the knee is close to full extension, the ankle in >10° dorsiflexion and the foot in external rotation.

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hamstring injuries to be present in team sports and to be rather common in individual sports such as track and field, dancing and deadlifting.[15 35 36] For instance, Askling *et al* described 18 injuries of the biceps femoris muscle in elite track and field athletes that occurred during competitive sprinting.[37] Among all muscle injuries included in this systematic review (both indirect and non-contact mechanisms), hamstring injuries were most frequently described as being running-related. This is in accordance with previous studies that reported more than half of imaging-confirmed hamstring injuries being related to running or sprinting.[38–40] The running phase in which athletes are most vulnerable to hamstring injury remains a matter of debate.[41] Most musculoskeletal modelling studies identified the open-chain late swing phase as being most vulnerable to injury.[42–44] During this phase of the gait cycle, the muscle-tendon unit of the biceps femoris lengthens, although the contractile elements themselves may potentially remain isometric.[45] This finding aligns with most studies in this systematic review that indicated the late swing phase as the injury-inciting phase of the gait cycle.[11–13 16 25 26 31 33] One of the included studies described the rapid transition from knee flexion to extension in the mid-swing phase as another critical aspect[34]; one was unable to investigate this aspect,[10] one did not analyse it[21] and one study included only deadlifting injuries. The authors, however, would like to highlight that most studies only speculated on the running phase.

Besides running, this systematic review identified further key underlying activities, for example, decelerations (including lunging or stopping manoeuvres with and without prior perturbation through indirect contact mechanism) and kicking. Joint movement direction varied in the hip, whereas the knee was extending in almost all cases. Importantly, some activities are rather sport-specific. For example, in waterskiing, the combination of a flexed hip and extended knees while experiencing a forward pull has been recognised as a common injury pattern leading to hamstring injury.[46] Historically, such injuries have been described as (over)stretching-related. The injury kinematics for these types of hamstring injury comprise a hip flexion position with an extended knee.[47]

Additionally, Kerin *et al* identified ipsilateral trunk flexion as a typical kinematic injury characteristic.[13] However, previous research rarely relied on imaging confirmation, nor did it consider injury classifications, such as the British Athletics Muscle Injury Classification.[48] For instance, it is possible that proximal hamstring avulsions have different injury patterns than hamstring injuries of the myotendinous junction. This systematic review of video-recorded hamstring injuries indicates that many hamstring injuries are due to sporting activities; however, a proportion of all hamstring injuries is caused by simple low-impact falls, slips and trips.[49] This has not been well described in the literature,[1] but one report of non-athletes described six patients with flexion of the hip and hyperextension of the knee during everyday activities such as missing a step during walking or gardening.[50]

Adductor and adductor longus injuries

Almost three-quarters of the adductor injuries were non-contact injuries, and one-quarter were indirect contact injuries. The injury-inciting activities of adductor muscle injury varied greatly and may include kicking, reaching, jumping, ball receiving, dribbling and change of direction. However, despite the variety of injury-inciting activities, most injury patterns were characterised by rapid muscle lengthening during activities that are usually characterised by high muscle activation.[10 11 14 20] Muscle-tendon unit lengthening is typically caused by hip extension, abduction and external rotation.[14 20] The combination of these specific injury pattern characteristics is rather limited to certain types of sports, such as football, ice hockey or basketball.[51 52] As there were only four studies investigating adductor injuries, data were limited overall, and further research on adductor injury mechanisms and kinematics is necessary to generate a profound understanding of injury causation.

Quadriceps and rectus femoris injuries

In quadriceps injuries, only one in 10 injuries was running-related. This finding is consistent with previous research describing kicking as a primary mechanism of quadriceps injuries in team sports.[11 19 53] As kicking starts with the backswing phase, the quadriceps needs to work eccentrically in order to control hip extension and knee flexion. Through the early forward swing phase (wind-up phase), the hip is flexing while the knee remains in a flexing position. The quadriceps is still contracting eccentrically to prevent excessive knee flexion.[19 53 54] It is hypothesised that rectus femoris injuries occur during this wind-up phase due to simultaneous length changes and force generation,[53] with the intent of moving from knee flexion to extension, which could be seen in the results of the present analysis. Of note, muscle tissue damage can also be caused by direct injury mechanisms (eg, contusions, lacerations); however, these were not investigated in this systematic review due to fundamental differences in injury aetiology, incidence and time of absence.[55]

Calf and gastrocnemius injuries

The most typical injury-inciting activity for calf and gastrocnemius injuries found in this systematic review was running. This finding is consistent with Green *et al* describing running-related activities as one of the primary injury patterns.[56] It is assumed that calf injuries occur during the stretch-shortening cycle in the stance phase of the running gait cycle when the muscle is shortening in length, and the muscle-tendon unit is lengthening and shortening due to tendon stretch and recoil.[56–58] Two gastrocnemius injuries were captured on camera in detail, and both gastrocnemius injuries were closed kinetic chain injuries. The detailed description of injury kinematics of these two cases highlights the importance of case reports for this specific research question, given that muscle injuries in a lab setting can be considered a very rare entity. At the assumed injury frame, the knee was in a slightly flexed position between 0° and 6°. The ankle was performing a dorsiflexion movement with a joint position of 10°−20° dorsiflexion at the assumed injury time. This joint position is very similar to the vulnerable injury situation associated with Achilles tendon ruptures,[59] indicating a shared injury pattern concerning joint kinematics. It is important to note that video analysis is typically based on more severe (gastrocnemius) injuries. Therefore, overuse aetiologies, as often proposed for soleus muscle injuries, are unlikely to be included in video analysis studies.[60]

Differences and similarities across injury locations

Differences were identified regarding the activities performed during the assumed injury-inciting activity. More than 50% of the reported hamstring and calf injuries (ie, lower limb functional posterior chain) were associated with running compared with less than a quarter for adductor and for quadriceps injuries. This result is consistent with the muscle requirements for running and sprint performance.[61] The hamstring muscles are needed for the swing phase during the running gait cycle, whereas the calf is essential for the specifics of the stance phase. Both muscle groups are processing high eccentric demands, which can contribute to injury. Kicking injuries were present in more than

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two-thirds of the quadriceps injuries, about a third in adductor injuries and less than a quarter of the hamstring injuries. All of the muscle groups are involved in the kicking manoeuvre but in different phases, which might lead to varying frequencies. The greatest elongation with simultaneous activation of the adductor longus occurs in the backswing phase of the kicking movement, whereas the rectus femoris generates high eccentric forces in the early swing phase of the kick. The hamstring muscles are at length towards the end of the kick when high muscle activation or overstretch could lead to muscle injury.[47 53 62] These results indicate the need for muscle-specific injury prevention strategies (eg, hamstring and calf injury prevention strategies may include preparation for running and sprinting, as well as acceleration and deceleration exercises).

Indirect and non-contact muscle injuries often occur due to overstretching or excessive strain on muscle fibres.[1] In addition, these injuries are associated with eccentric movements, leading to the muscle-tendon unit being overstretched while simultaneously generating force.[1] The majority of muscle injuries were reported in biarticular muscle groups. As biarticular muscles are subjected to high neuromuscular demands during different phases of movement and their multifunctional role in movement control, they are at an increased risk of injury.[5] Regarding biarticular muscles, the rate of non-contact muscle injuries ranged between three-quarters and nine in 10 injuries. This constant result provides relevant information to help the development of muscle injury prevention strategies, which, whatever the muscle group, could be targeted in prioritising non-contact injury mechanisms and to prepare these actions with high loads and implement specific exercises for the demand profile of specific sports activities. However, understanding the indirect contact mechanism of muscle injuries can potentially include valuable sport-specific solutions to risk mitigation programmes (eg, including perturbation-type exercises of deceleration activities of lunging, stopping or landing). Despite available prevention programmes, muscle injury rates have not decreased in recent decades.[63] Recently, previous assumptions and dogmas have already been challenged.[41] To improve patient care, it seems important to correlate theoretical concepts and modelling studies with real-world data from video analysis of injury. Video analysis of muscle injuries may expand the understanding of injury causation, among others, by considering the complex interplay of contributing factors such as situational characteristics, lower leg anatomy and neuromuscular control. For future research, a major advancement could lie in the assessment of (magnetic resonance) imaging in order to correlate situational characteristics of injury with specific imaging findings (eg, intramuscular injuries, myotendinous junction injuries or bony avulsions).

The timing of injury during team sports matches remains a matter of debate. Some previous research found more injuries towards the second half,[64 65] but other studies reported more injuries in the first half.[11 12 14] Our systematic review of six studies showed a slight tendency towards more injuries in the first half. Practically, this may lead to the assumption that the sudden onset of high-intensity activities (eg, by insufficient warm-up or by selecting players with a pre-existing minor muscle injury) could have a higher impact on muscle injuries than fatigue at the end of a match. However, in the female subgroup, limited data were available, showing an even distribution of injuries in the first and second half. Due to limited data, it is necessary to do further research on this subject. Of note, the match time is of less interest than the minute the player is on the field (in case of substitutions). At present, no conclusions on the impact of match time on injury occurrence can be drawn.

Practical implications for injury risk reduction

A causal understanding of injury occurrence is essential for designing injury risk reduction strategies.[6] The findings from our systematic review help guide the development of new injury prevention programmes. Many researchers and clinicians suggest that injury risk can be reduced through exercise programmes that specifically target the underlying causes of injury ('the problem is the solution').[1 66 67] For instance, since hamstring injuries are thought to occur primarily during sprinting and eccentric muscle contraction, sprinting itself and the Nordic hamstring exercise have been widely implemented in clinical practice.[67–70] Studies indicate that specific exercise-based prevention programmes can reduce hamstring injury rates by up to 35%.[71] Based on this review, strategies to prevent hamstring (and also calf) injuries may include preparation for running and sprinting, as well as acceleration and deceleration (including kicking and reaching) drills. Perturbation-based training may also be beneficial, given the neuromuscular component, especially in indirect contact mechanism of injury occurrence. In contrast, adductor injuries, which often occur when the muscle is at maximal length, may be better addressed through exercises involving hip external rotation, hip abduction and hip extension. Additionally, muscle activation demands should be incorporated. Given the preliminary findings of sport-specific injury patterns, tailored prevention programmes need to be considered for individual sports. The authors emphasise, however, that, while highlighting the aforementioned translation of findings to clinical practice, the effectiveness of preventive approaches must ultimately be proven with clinical trials.

Limitations

A major methodological consideration of this systematic review is that most of the studies were focused on muscle injuries in football. The large number of running-related injuries would potentially change if more data from other sports were available (eg, gymnastics). In addition, data on female muscle injuries was rarely available due to the lack of video recordings in female sports. The authors call for prioritising future research to close this data gap. Most studies analysed hamstring injuries; limited data were available for other muscle groups. Furthermore, some studies failed to adequately report information about the video source and quality, the raters' qualification, a representative study population and validated methods. This led to varying quality scores from low to high quality, which might have influenced further analysis. Reporting can be improved by adhering to specific guidelines for conducting and reporting studies, which could be based on the QA-SIVAS scale.[8] The studies used different methods and terminology, and some had limited data available, which altogether might have led to the varying heterogeneity in the meta-analyses. Sensitivity analysis revealed lower heterogeneity by excluding studies of lower quality. This may indicate that future research needs to be guided by rigorous quality standards. Another limitation of this study is the aspect of video analysis itself. Video analyses can only be as good as the video quality (including recording procedures and standards) and rely on multiple camera views to enable high-quality kinematic analysis. To advance research and to standardise the documentation of injury patterns, collaborative efforts aimed at achieving international consensus are needed. This would facilitate further comparative studies and meta-analyses. Efforts have been made recently regarding terminology used in football and netball, but standardisation remains insufficient.[72 73]

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CONCLUSION

This systematic review and meta-analysis of 728 video-detected cases identified distinct patterns associated with muscle injuries. Most muscle injuries were non-contact, and nearly half of all muscle injuries were related to running. The remaining injuries typically occurred during sport-specific manoeuvres that involved rapid changes in muscle-tendon unit length and simultaneous muscle contraction. Future research may now focus on raising quality standards of video recording procedures, better acknowledging and prioritising less well-studied populations (eg, female athletes, track and field athletes), investigating rare injury locations and translating research findings into clinical practice.

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